

CLOUD COMPUTING ASSIGNMENT

CREATE AND CONFIGURE CLOUD STORAGE USING AWS S3

Deliverable: Bucket setup with example files uploaded and access permissions configured

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1. Introduction

Amazon Simple Storage Service (Amazon S3) is an object storage service used to store files as objects inside buckets. In this practical, an S3 bucket named “omkar-cloud-storage” was created in the Europe (Stockholm) region (eu-north-1). Example files were uploaded to the bucket, and access permissions were configured using S3 bucket and object access controls.

2. Objective

To create and configure an Amazon S3 bucket, upload example files, configure access permissions, and verify that an uploaded object can be accessed using its S3 object URL.

3. AWS S3 Configuration Summary

Configuration Item	Observed Configuration
Cloud service	Amazon S3
Bucket name	omkar-cloud-storage
AWS Region	Europe (Stockholm) – eu-north-1
Bucket objects	3 objects: img.avif, img2.avif, wb1.html
Bucket versioning	Disabled
Bucket ABAC	Disabled
Default encryption	Server-side encryption with Amazon S3 managed keys (SSE-S3)
Block Public Access	Off
Bucket policy	Allows s3:GetObject for the bucket objects with Principal “*”

4. Procedure and Evidence

4.1 Verify the Created S3 Bucket

The Amazon S3 Buckets page shows one general-purpose bucket named “omkar-cloud-storage”. The bucket is listed in the Europe (Stockholm) region, eu-north-1. This screenshot provides direct evidence that the required S3 bucket was created successfully.

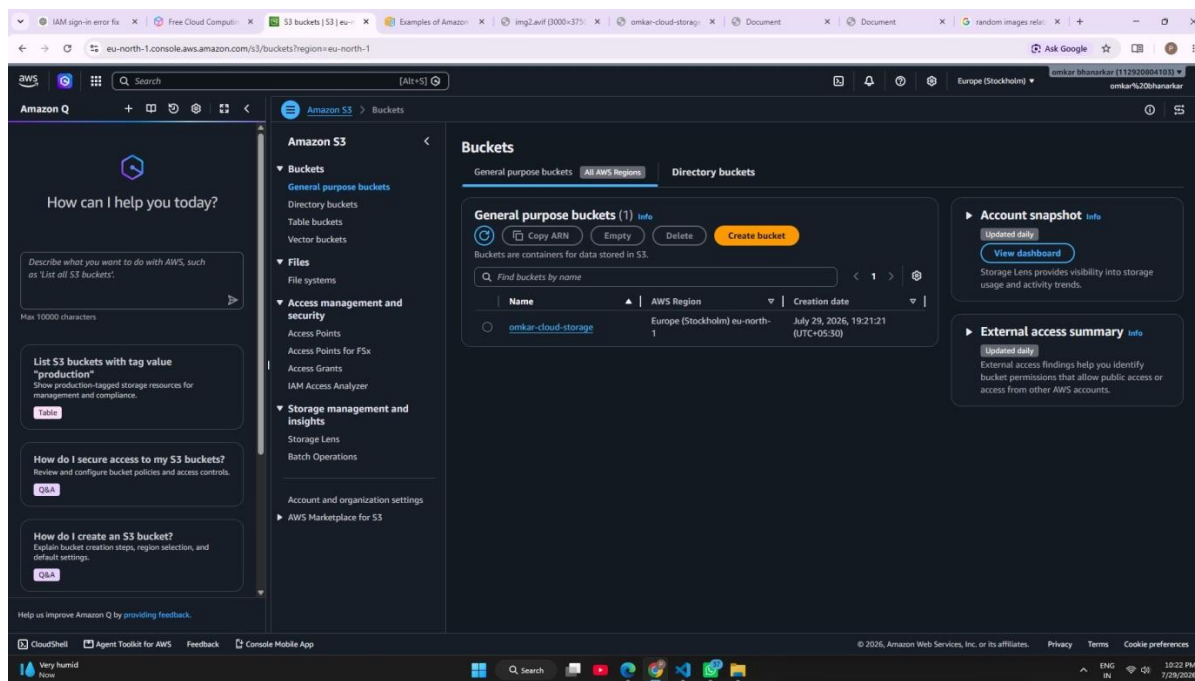


Figure 1: Amazon S3 Buckets page showing the created bucket “omkar-cloud-storage” in Europe (Stockholm), eu-north-1.

4.2 Review Bucket Properties

The bucket Properties page displays the bucket overview, including the AWS Region, Amazon Resource Name (ARN), and creation date. It also shows that Bucket Versioning is disabled and Bucket ABAC is disabled. These details confirm the basic configuration of the created bucket.

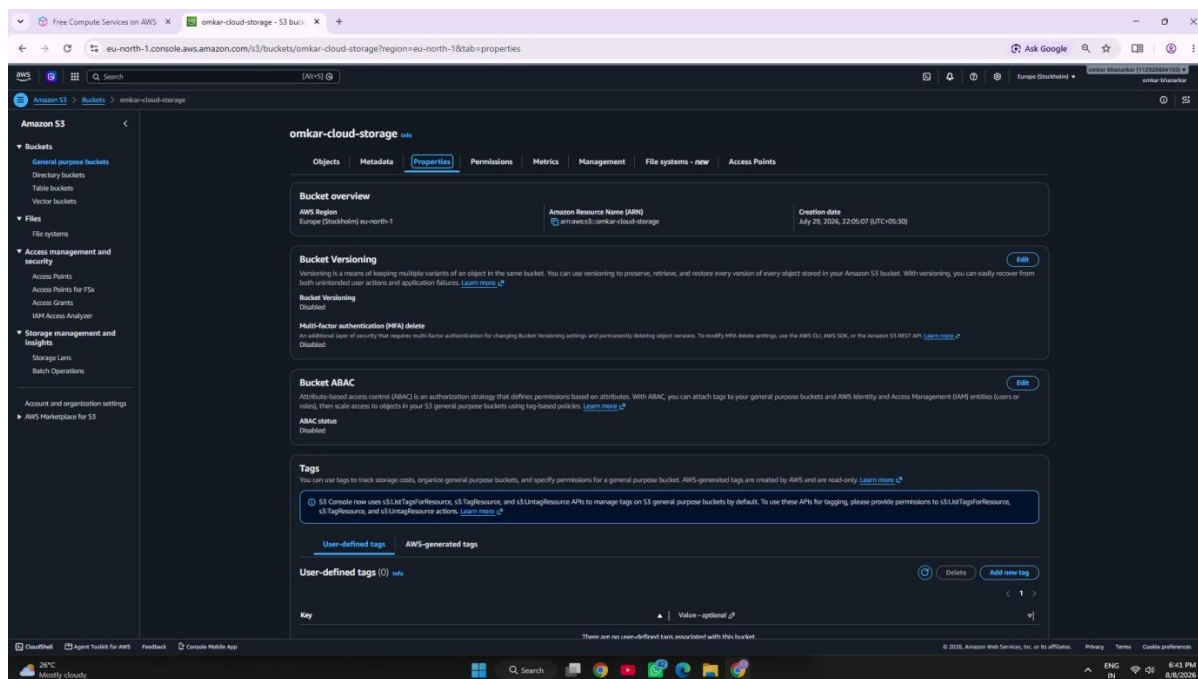


Figure 2: S3 bucket Properties page showing the bucket overview, region, versioning, ABAC status, and tags.

4.3 Review Storage and Encryption Properties

The lower portion of the Properties page shows Default encryption. The screenshot indicates Server-side encryption with Amazon S3 managed keys (SSE-S3) for new objects stored in the bucket. It also shows that no Intelligent-Tiering Archive configurations or CloudWatch log deliveries are configured.

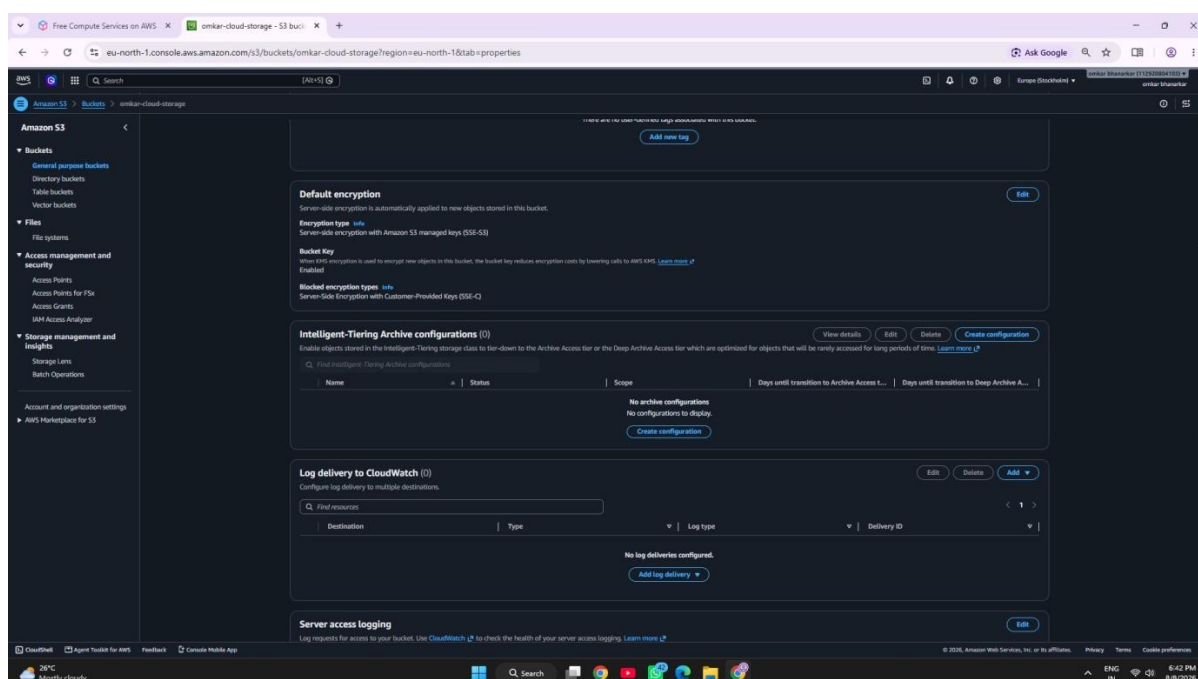


Figure 3: S3 bucket Properties page showing default encryption using SSE-S3 and related storage/logging settings.

4.4 Upload and Verify Example Files

The Objects page shows three objects stored in the bucket: “img.avif”, “img2.avif”, and “wb1.html”. Their file types, last-modified timestamps, sizes, and storage class are visible. This directly satisfies the requirement to upload example files to the cloud storage bucket.

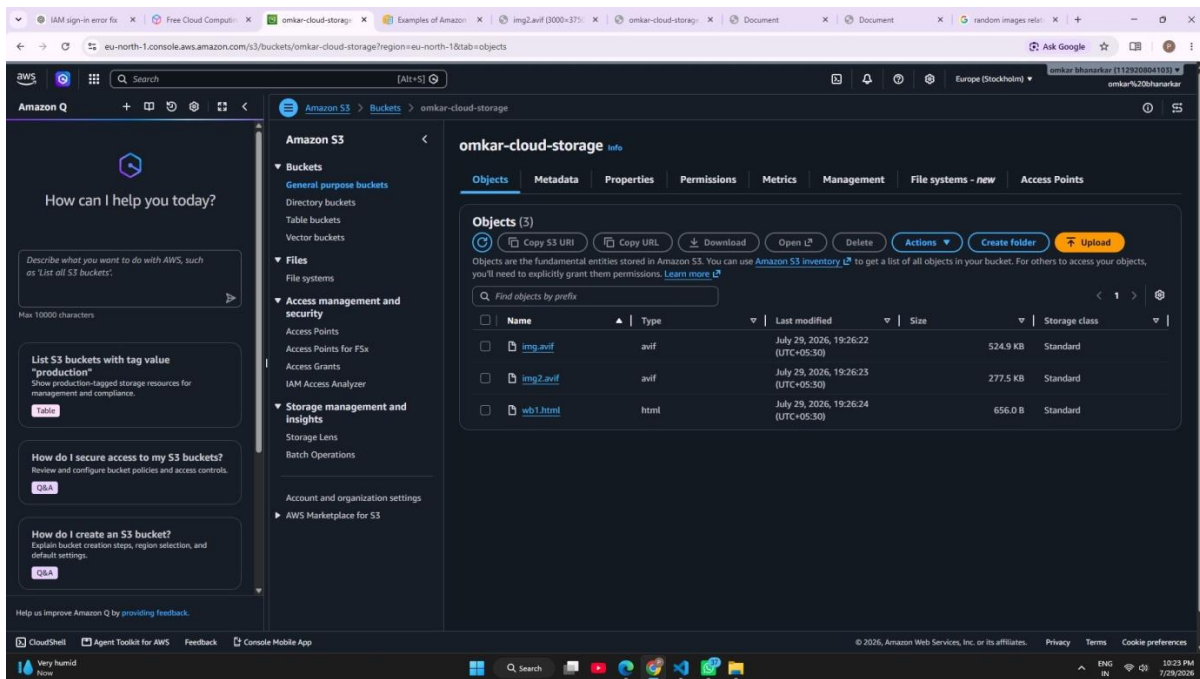


Figure 4: S3 Objects page showing the three uploaded example files: img.avif, img2.avif, and wb1.html.

4.5 Configure and Review Object-Level Permissions

The Permissions page for the object “img.avif” shows its Access Control List (ACL). The bucket uses the bucket-owner-enforced Object Ownership setting, so bucket policies are used to control access. The object owner has read access to the object and read/write access to the Object ACL, while the public and authenticated-user ACL entries do not grant permissions.

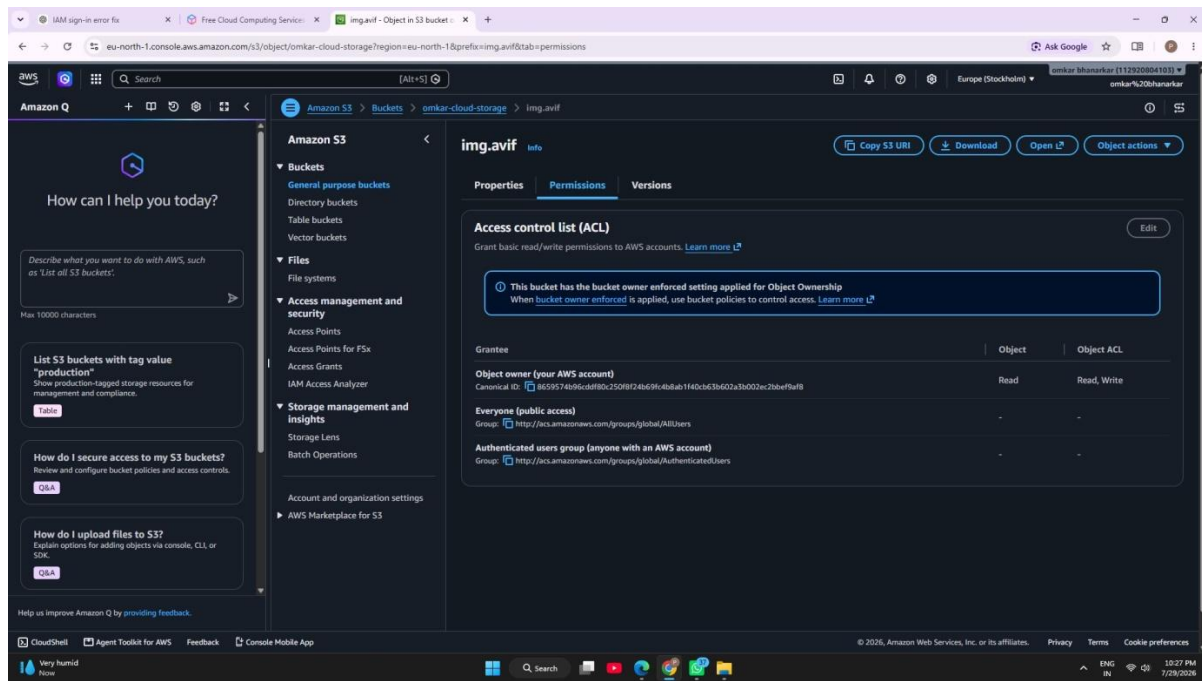


Figure 5: Permissions page for “img.avif” showing Object Ownership and the object Access Control List (ACL).

4.6 Configure and Review Bucket-Level Access

The bucket Permissions page shows that Block All Public Access is turned Off. The same page contains a bucket policy with Principal “*”, Effect “Allow”, and Action “s3:GetObject” for resources under the bucket. Therefore, the configured policy provides read access to bucket objects through the object URL. This is the main screenshot demonstrating the bucket-level access permission used in this practical.

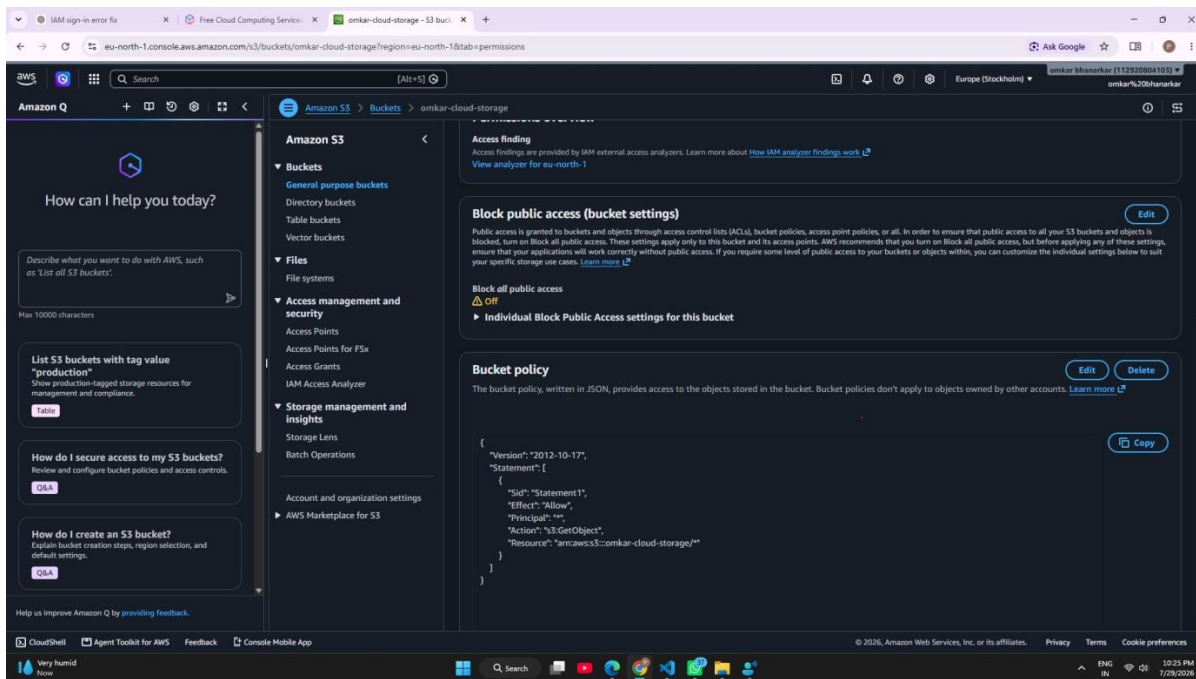


Figure 6: S3 bucket Permissions page showing Block Public Access set to Off and a bucket policy allowing s3:GetObject.

4.7 Verify Object Accessibility

The final screenshot shows “img.avif” opened directly through the S3 object URL. The image is displayed successfully in the browser. This verifies that the access configuration allows the uploaded object to be retrieved through its S3 URL. (OBJECT URL :: <https://omkar-cloud-storage.s3.eu-north-1.amazonaws.com/img.avif>)

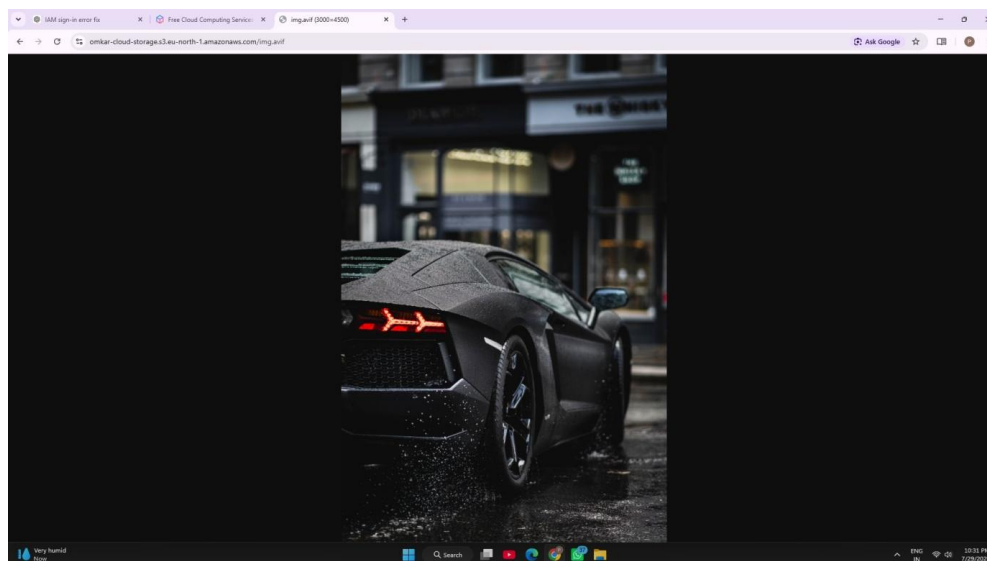


Figure 7: Uploaded “img.avif” object successfully opened through its S3 object URL.

5. Access Permission Analysis

The screenshots show two related access-control mechanisms. At the object level, the ACL is not being used to provide public access because the bucket owner-enforced Object Ownership setting is enabled. At the bucket level, Block Public Access is turned off and the bucket policy explicitly allows the s3:GetObject action to Principal “*”. Consequently, the object can be retrieved through its S3 URL, as demonstrated in Figure 7.

For a production environment, public read access should only be enabled when it is specifically required. Otherwise, Block Public Access should normally remain enabled and access should be granted only to the required users, roles, or services. For this practical, the screenshots document the access configuration that was actually implemented.

6. Verification of Deliverable

Requirement	Evidence	Status
Bucket setup	Figure 1 and Figures 2–3 show the created bucket and its properties.	Completed
Example files uploaded	Figure 4 shows three objects in the bucket.	Completed
Access permissions configured	Figures 5–6 show object ACL and bucket-level permissions/policy.	Completed
Access verified	Figure 7 shows the uploaded object opened from its S3 URL.	Completed

7. Result

The AWS S3 cloud storage task was successfully completed. The bucket “omkar-cloud-storage” was created in the Europe (Stockholm) region, three example objects were uploaded, access permissions were configured through S3 access-control settings and a bucket policy, and the uploaded image object was successfully accessed through its S3 object URL.

8. Conclusion

This practical demonstrated the complete basic workflow of Amazon S3 cloud storage: creating a bucket, reviewing its configuration, uploading objects, configuring access permissions, and verifying object access. The included screenshots provide evidence for each major part of the required deliverable.

9. Submission Checklist

- ☑ Bucket created: omkar-cloud-storage.
- ☑ Example files uploaded: img.avif, img2.avif, and wb1.html.
- ☑ Bucket properties and encryption settings documented.
- ☑ Object-level ACL and bucket-level access permissions documented.
- ☑ Object access verified using the S3 object URL.
- ☑ Report contains screenshots, explanations, result, and conclusion.